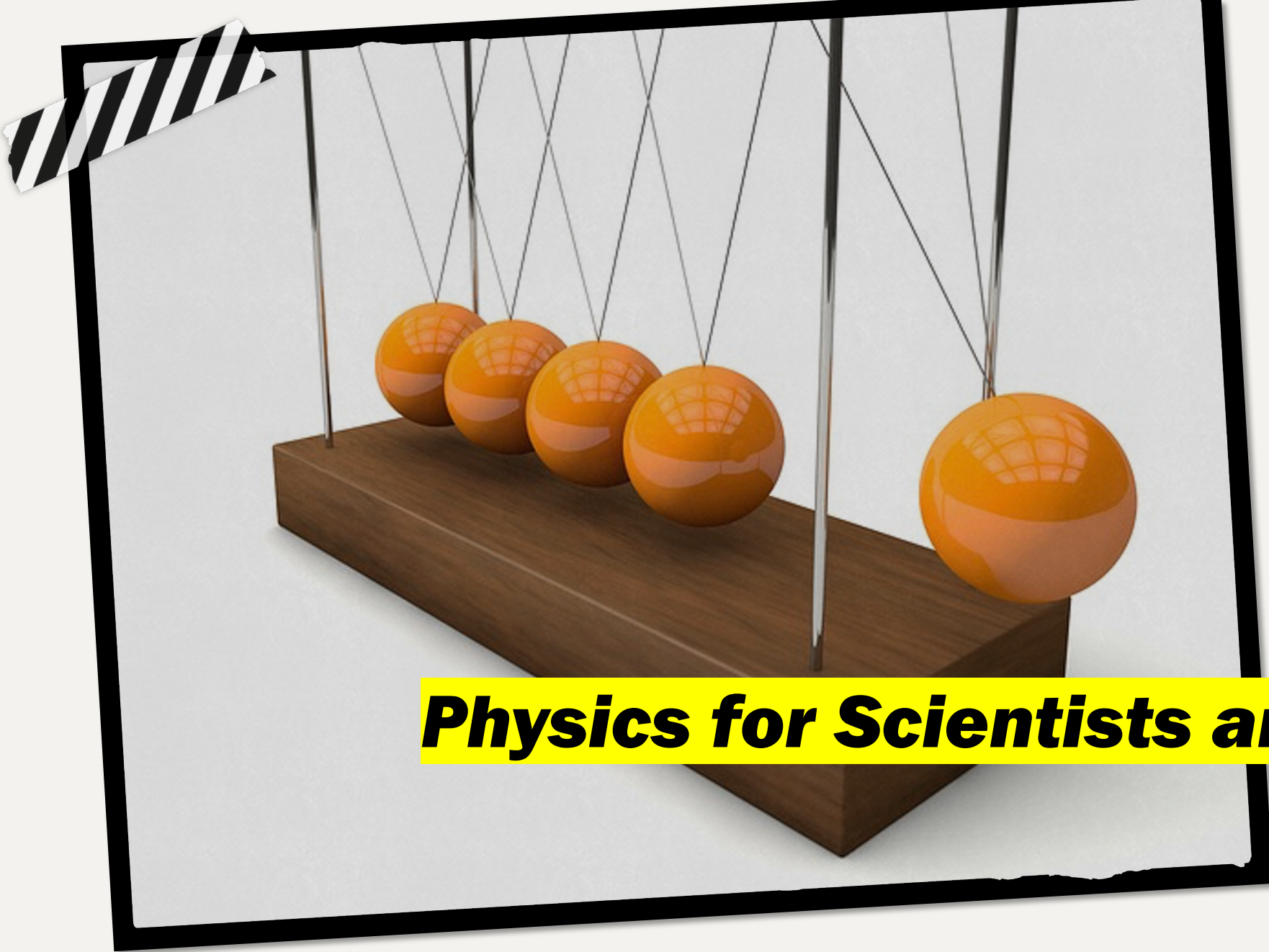




PHYS 151

Physics for Scientists and Engineers

Aug 26th, 2026



Lecture instructor

Dr. Jiang-Xiazi Lin (jiangxiazi.lin@emory.edu)

Assistant professor of physics

Physics mentors (Learning Assistants)

Ava Tholen, Andrew Duh, and Aishani Valluru

Lab supervisor

Cory Donofrio (cjdonof@emory.edu) ** All lab-related questions please contact Cory.

Teaching Assistants

Robert Sevik, Tai Do, Antoine Wang, Venkatesh Jha

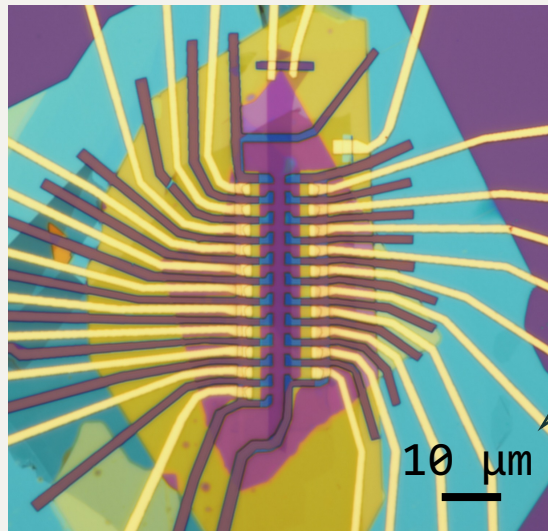


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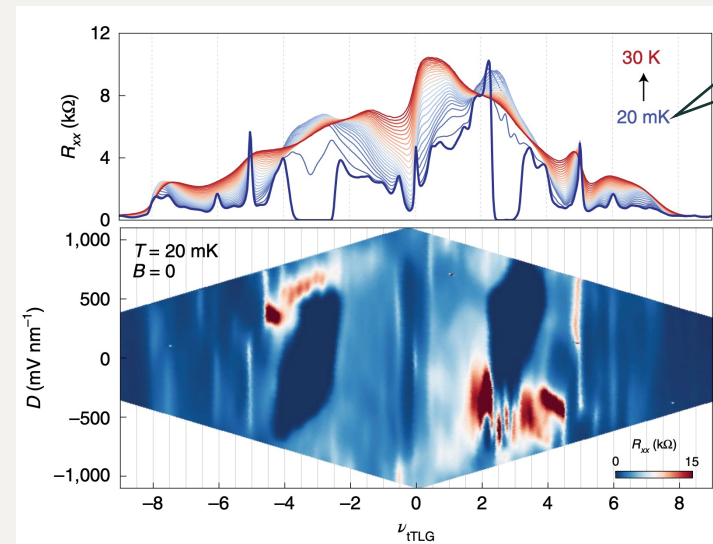
Assistant professor of physics

Research: experimental condensed matter physics



J.-X. Lin, et al., *Science* (2019)

Micro device



J.-X. Lin, et al., *Nature Physics* (2019)

Superconductivity

That's cold!



Spin coater



About this course

- Textbook: Physics for Scientists and Engineers, 10th Edition, part 1,2,3
- Weekly schedule:

Lectures	Mon Wed 1:00pm - 2:15pm. Math & Science Center E208
Office hours	Tue 4-5pm and Thu 12-1pm. Math & Science Center N246
Mentor sessions	TBD
Lab	See a separate Canvas page

Homework:

- Post after each Wednesday class, due before the next Wednesday class.
 - 1st homework assignment will be posted today.
 - WebAssign class key: **emory 6650 3919**
- Exams: Two mid-term exams and one final exam (Dec 16th)
 - For more info, please read the **syllabus** on Canvas!
 - Please turn on Canvas announcement notification (if not already on).



Physics and math

- Physics is about using mathematical models to understand and summarize the patterns of natural phenomena.
- We will use calculus in this course (e.g., trigonometric functions, derivative and integral of polynomial functions, etc.)
- Math reference: appendix B at the back of the textbook

The image shows a chalkboard with handwritten mathematical derivations. On the left, a graph of a function $y = g(x)$ is shown with a secant line and a tangent line. The secant line connects points $(x, g(x))$ and $(x+h, g(x+h))$. The slope of the secant line is labeled as $\frac{g(x+h) - g(x)}{(x+h) - x}$. The tangent line is labeled as $\frac{dy}{dx}$. The slope of the tangent line is labeled as $\frac{dy}{dx}$. The slope of the secant line is labeled as $\frac{g(x+h) - g(x)}{(x+h) - x}$. The slope of the tangent line is labeled as $\frac{dy}{dx}$.

Find the derivative

$$f(x) = x^2$$
$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$
$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h}$$
$$f'(x) = \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - x^2}{h}$$
$$f'(x) = \lim_{h \rightarrow 0} \frac{2xh + h^2}{h}$$
$$f'(x) = \lim_{h \rightarrow 0} (2x + h)$$
$$f'(x) = 2x$$

Slope of $f(x) = \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{(x+h) - x}$

$$\frac{df}{dx} = \frac{d}{dx} (x^n) = nx^{n-1}$$
$$\frac{df}{dx} = \frac{d}{dx} (x^2) = 2x$$
$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$$
$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$



Learning advices

- **Lectures**

We have **A LOT** to cover. Lectures will focus on derivations and explanation of concepts (~2/3), applications or examples (~1/3). Take your own notes even though my notes will be posted.

- **Homework**

Thinking is a VERY IMPORTANT part of learning physics.

** Suggestion: use symbols during analysis, only plug in the numbers in the end.

** If you must ask AI for help? 1) Work on it all by yourself! 2) If you are stuck, ask AI “how” “why use this” “why not that”, dig until everything makes sense and never walk away with any confusion. Don’t skip step one. Remember AI makes mistakes (quite often), and AI can’t learn for you.

What if you still have questions?

- **Office hours and mentor sessions**

Bring your questions, or just be there listening.

- **Lab**

Managed by Cory.



Learning advices

- **Exams**

- Variations of lecture examples and homework problems will appear in exams!
- Closed book. Equation sheet given.
- Only symbolic. No calculator needed or allowed.
- Practice exams will be given.

Build your physics mindset, train the way of thinking like a scientist!

